

International Journal of Digital Earth

ISSN: 1753-8947 (Print) 1753-8955 (Online) Journal homepage: www.tandfonline.com/journals/tjde20

Investigation of the spatio-temporal transferability of large-scale random forest crop classification models and potential influencing factors

Shuai Xie, Naijie Zhang, Liangyun Liu & Lin Sun

To cite this article: Shuai Xie, Naijie Zhang, Liangyun Liu & Lin Sun (2025) Investigation of the spatio-temporal transferability of large-scale random forest crop classification models and potential influencing factors, International Journal of Digital Earth, 18:2, 2563211, DOI: 10.1080/17538947.2025.2563211

To link to this article: https://doi.org/10.1080/17538947.2025.2563211

© 2025 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group

View supplementary material

Published online: 29 Sep 2025.

Submit your article to this journal

Article views: 1096

View related articles

View Crossmark data

Citing articles: 4 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=tjde20

INTERNATIONAL JOURNAL OF DIGITAL EARTH
2025, VOL. 18, NO. 2, 2563211
https://doi.org/10.1080/17538947.2025.2563211

OPEN ACCESS Check for updates

Investigation of the spatio-temporal transferability of large-scale random forest crop classification models and potential influencing factors

Shuai Xie[a], Naijie Zhang[a], Liangyun Liu[b] and Lin Sun[c]

[a]School of Information and Control Engineering, Qingdao University of Technology, Qingdao, People's Republic of China; [b]Key Laboratory of Digital Earth Science, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, People's Republic of China; [c]College of Geodesy and Geomatics, Shandong University of Science and Technology, Qingdao, People's Republic of China

ABSTRACT

Training samples are crucial for remote-sensing-based crop classification; however, field surveys for data collection over large areas are costly and challenging. Model transfer offers a viable solution. This study evaluates the spatio-temporal transferability of machine learning models for large-scale crop classification and examines factors influencing classification performance across different training regions and years. A Random Forest model trained on Landsat/Sentinel-2 time series and CDL crop labels from three U.S. states was applied to six European countries for crop mapping. Results showed that models trained in Kansas achieved high overall accuracy (88.1–96.8%), with minimal accuracy variation (0.6–2.1%) across training years, demonstrating strong adaptability. However, model performance varied significantly across training regions (up to 21.6% accuracy difference), with Kansas outperforming Minnesota and South Dakota. Interestingly, models trained on merged regions or years performed worse than the best individual models, indicating that broader spatial or temporal training data cannot guarantee better transferability. Transferred models achieved accuracy comparable to locally trained models (≤2.0% difference). Temperature similarity between regions is a key factor, with greater annual temperature alignment leading to better transferability. This study offer a practical solution to large-area crop classification without local sample and guide the selection of optimal training areas.

ARTICLE HISTORY
Received 17 March 2025
Accepted 25 August 2025

KEYWORDS
Large-area; crop classification; spatio-temporal transferability; annual temperature pattern; google earth engine

1. Introduction

Accurate crop type maps are a valuable and essential source of information for many applications, including food security, crop rotation, crop yield prediction, and disaster prevention (Wardlow and Egbert 2008; You et al. 2014). Remotely sensed data is an important data source for obtaining large-scale crop information, and supervised machine learning methods combined with satellite remote sensing imagery are widely used for large-area crop classification (Griffiths, Nendel, and Hostert 2019; Inglada et al. 2017; Preidl, Lange, and Doktor 2020). To obtain highly accurate crop type maps, supervised machine learning algorithms require a large number of widely distributed training samples. Crop type samples are usually obtained through ground surveys; however, the sample collection process suffers from limitations, such as high cost, inefficiency, and inaccessible locations for remote areas (Antonijević et al. 2023; Orynbaikyzy, Gessner, and Conrad 2022). Therefore, the main difficulty in remote-sensing-based crop classification over large areas lies in the acquisition of training samples. A feasible way to solve this problem is to utilize the transferability of the classification model. This kind of method first trains a crop classification model in the source domain and then applies the trained model to the target domain for crop classification. The source and target domains here respectively refer to the training areas that already have a large amount of high-quality ground-labeled data and the mapping areas with no ground samples.

CONTACT Liangyun Liu liuly@radi.ac.cn Key Laboratory of Digital Earth Science, Aerospace Information Research Institute, Chinese Academy of Sciences, No. 9 Dengzhuang South Road, Haidian District, Beijing 100094, People's Republic of China
Supplemental data for this article can be accessed online at https://doi.org/10.1080/17538947.2025.2563211.

© 2025 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group
This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial License (http://creativecommons.org/licenses/by-nc/4.0/), which permits unrestricted non-commercial use, distribution, and reproduction in any medium, provided the original work is properly cited. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

In recent years, the transferability of classification models has been widely used in studies of crop type mapping for areas without samples. Wang et al. proposed a crop type mapping method without field labeling data based on random forest transfer and unsupervised clustering techniques. The spatial and temporal transferability of RF classification model across states in the Midwestern United States was investigated using Cropland Data Layer (CDL) crop type labeling data and Landsat satellite imagery data. RF models trained in regions and years with similar growing seasons were able to achieve 80% crop classification accuracy when transferred to the target region (Wang, Azzari, and Lobell 2019). Hao et al. proposed a migration-model – based crop classification method using US CDL data as training samples. The accuracies of the RF crop classification model trained in the US region were applied to other regions (i.e. Hengshui, China; Alberta, Canada; and Nebraska, USA), and the results showed that the overall classification accuracy of the migration model was more than 86% in all three study areas (Hao et al. 2020). Luo et al. developed a high-resolution crop mapping method based on the migration of classification models from the limited European areas. The 10 m crop type maps for 10 other EU countries were generated using crop classification models trained in the UK and France, with an overall accuracy of more than 89% and an R^2 of more than 0.9 with respect to official crop statistics (Luo et al. 2022). Orynbaikyzy et al. evaluated the spatial transferability of RF crop classification model based on Sentinel-1 and Sentinel-2 data using seven regions distributed in Germany as the study area, and analyzed the effects of optical and radar data as well as the selection of spatial features on the classification accuracy of the transferred model. The results showed that the combination of optical and radar data outperformed single sensors, while single radar data exhibited the lowest loss of accuracy; furthermore, compared to the full feature set, spatial feature selection was able to significantly reduce the number of input features and maintain good prediction performance for the target domain (Orynbaikyzy, Gessner, and Conrad 2022). Zhang et al. proposed an automatic crop type mapping method based on spatio-temporal crop information and Sentinel-2 data. The proposed method extracted high-confidence crop labels from historical CDL crop distribution maps for classification model training, and then automatically performed in-season crop mapping. This study conducted a series of experiments using the Mississippi Delta as the study area, and the results showed that the proposed method can achieve 80% to 90% agreement with CDL products (Zhang et al. 2022). A generalized model was proposed by Qadir et al. for sunflower area mapping based on Sentinel-1 data, phenological metrics and RF classifier. Experiments on several study sites in Europe and the US have shown the promising spatio-temporal generalization of the developed model (Qadir et al. 2024). In addition to machine learning models, several deep learning structures, such as long short-term memory (LSTM) (Xu et al. 2020), U-Net (Ge et al. 2021) and deep adaptation network (DAN) (Wang et al. 2023), have recently been developed for crop type mapping and have proven to be highly spatio-temporally transferable.

Machine and deep learning models have been shown to have good spatial and temporal generalization for crop classification. However, the current study has two limitations. First, the selected study area was usually not large. The spatio-temporal transferability of crop classification models for large areas (e.g. at the national scale), where crop types have more varied growing environments and more complex spectral characteristics, needs to be further investigated. Second, crop classification models obtained from different training areas usually have diverse transferable performances, but the factors contributing to this have rarely been quantitatively analyzed and explored. Building on this, the objective of this study was to explore the spatio-temporal transferability of random forest crop classification models over large areas at the national scale across continents, and quantitatively analyze the potential factors influencing transferability performance. The novel contribution of our work is that we offer deeper insights into the effectiveness of cross-regional and cross-temporal large-scale crop classification models, and provide a quantitative analysis method guiding the selection of optimal training areas for model transfer.

We performed satellite data preprocessing, time series construction, feature extraction, and crop mapping based on Google Earth Engine (GEE) platform, since GEE can provide the required geospatial data archives, functions and machine learning algorithms that are readily accessible, as well as cloud computing services, facilitating the development, application, and validation of crop classification models for large areas. The remainder of this paper is organized as follows. Section 2.1 introduces the study areas and the data we used. How we develop the crop classification model is described in Section 2.2. Section 3 presents the quantitative evaluation of the transferred models from different training areas and years, the comparison with local models and the factors affecting the spatio-temporal transferability of the models. The results are discussed in Section 4 and the conclusions of this study are given in Section 5.

2. Materials and methods

2.1. Study area and data

2.1.1. Study area

In this study, corn and wheat, which are among the major global food crops, were selected as research targets. The United States has a wealth of data on crop types. According to the USDA National Agricultural Statistics Service website (USDA-NASS n.d. – d), Kansas, South Dakota, and Minnesota in the United States were chosen as training areas for crop classification as these three states have large planting areas for corn and wheat. These states are located in the central and north-central United States and cover 0.21, 0.20, and 0.23 million square kilometers, respectively, spanning longitudes from 89° to 104°W and latitudes from 37° to 49°N within humid continental and humid subtropical climate zones. In addition, according to the European Agricultural Statistics website (Eurostat n.d.), six countries with large planting areas for corn and wheat in Europe (Germany, France, Romania, Poland, Spain, and Italy) were selected as the mapping areas for crops. The areas of these countries are 0.36, 0.54, 0.24, 0.31, 0.51 and 0.30 million square kilometers, respectively. Compared to the regions situated in the interior of the United States, these European countries are adjacent to the Atlantic Ocean and Mediterranean Sea. Thus, the climate in Europe is predominantly temperate and continental, but has an oceanic climate along the western coast and a Mediterranean climate in the southern regions. The geographical locations of the training and mapping areas used in this study were shown in Figure 1.

2.1.2. Satellite data

In this study, all available Landsat-8 Collection 2 Surface Reflectance (SR) and Sentinel-2 Level-2A (L2A) images covering the training and mapping areas were acquired based on the Google Earth Engine (GEE) cloud computing platform. The Landsat-8 Collection 2 SR products were newly released by the U.S. Geological Survey (USGS) (Crawford et al. 2023) and were generated using the LaSRC atmospheric correction algorithm (Vermote et al. 2016). Sentinel-2 L2A was provided by the European Space Agency (ESA) and was also an atmospherically corrected surface reflectance product generated using the sen2cor algorithm (Main-Knorn et al. 2017). Atmospheric correction can eliminate the interference such as atmospheric scattering and absorption, improve the accuracy of surface reflectance, and enhance the comparability and consistency of images across time and between sensors. As Landsat-8 and Sentinel-2 data have similar spatial resolution and spectral band configurations, six matched spectral bands in terms of wavelength and spectral response function were selected for the combined use of the two satellites, that is, Landsat-8 B2, B3, B4, B5, B6 and B7 bands (30 m for all six bands) and Sentinel-2 B2, B3, B4, B8A, B11 and B12 bands (10 m for the first three bands and 20 m for the last three bands). Sentinel-2 B8A instead of B8 was used for NIR region as B8A has a wavelength range and spectral response function more similar to its Landsat-8 counterpart, that is, the B5 band (Chastain et al. 2019; Xie et al. 2022). Each band of Sentinel-2 was resampled to 30 m, which is consistent with Landsat-8. The combination of Landsat-8 and Sentinel-2 can provide denser temporal observations than single-sensor data to construct accurate time-series feature representations and crop classification models (Liu et al. 2023). The number of Landsat-8 and Sentinel-2 images used for each study area for each year were shown in Table 1. Noted that we did not use images from 2018 as training

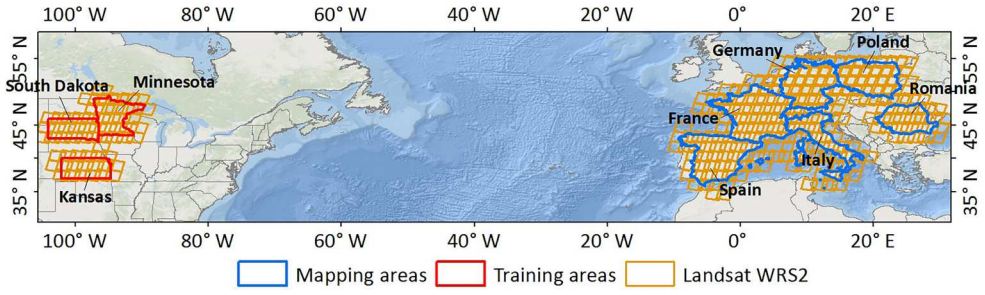


Figure 1. Geographic locations of the training areas and mapping areas used in this study.

Table 1. Sum of the number of Landsat-8 and Sentinel-2 images for each study area used in this study.

	Training areas			Mapping areas					
	Kansas	South Dakota	Minnesota	Germany	France	Romania	Poland	Spain	Italy
2018	–	–	–	12590	15536	6820	9494	16093	11935
2019	6333	5498	6984	–	–	–	–	–	–
2020	6385	5565	6966	–	–	–	–	–	–
2021	6364	5461	6950	–	–	–	–	–	–
2022	6367	5523	6962	–	–	–	–	–	–
2023	6344	5500	6972	–	–	–	–	–	–

data because, compared to 2019 and subsequent years, there were insufficient Sentinel-2 images from 2018 for the US region to construct robust time-series spectral features used for classification.

2.1.3. Meteorological data

Meteorological data, including air temperature and precipitation, were used to measure the similarity of climatic environments between regions and to quantitatively evaluate the factors influencing the performance of crop classification models obtained from different regions. Monthly air temperature and precipitation data for our study areas were collected from ERA5-Land Monthly Aggregated dataset (Muñoz Sabater 2019) in GEE. This dataset was generated by the Copernicus Climate Change Service (C3S) Climate Data Store (CDS), which provides global coverage of 50 climate variables at a spatial resolution of 10 km on a monthly basis from 1950 to the present. Two bands, named 'temperature_2m' and 'total_precipitation_sum', were selected from this asset. The band of 'temperature_2m' described the temperature of air in kelvin at 2 meters above the surface of land, sea or in-land waters. The 'total_precipitation_sum' band recorded the accumulated liquid and frozen water, including rain and snow, which falls to the Earth's surface, with the units of depth in meters. Monthly time series from January to December for each of the two bands were used to characterize the air temperature and precipitation of the study areas where crops were grown.

2.1.4. Crop type training and validation data

The crop training data used in this study were obtained from the Cropland Data Layer (CDL) dataset of the United States. This dataset, produced by the U.S. Department of Agriculture (USDA) (Boryan et al. 2011), utilized Landsat satellite imagery and extensive ground survey data to generate annual time-series crop type maps for the entire United States since 2008. Validation of the CDL product has shown a mapping accuracy of approximately 95% for the major crop types (i.e. corn, wheat, and soybeans) (Boryan, Yang, and Di 2012). CDL data were widely used as label data to satellite imagery for land use/cover classification (Yan and Roy 2015; Zhai, Qu, and Hao 2018), crop identification in early and within growing season (Zhang et al. 2022), and crop classification model transfer (Hao et al. 2020; Wang, Azzari, and Lobell 2019). In addition, the accuracies of the crop mapping results generated in this study were quantitatively evaluated using the validation samples from European LUCAS in-situ data (d'Andrimont et al. 2020). LUCAS was a ground survey that collected information on land cover, land use, and crop types across the European Union. This effort was conducted every three years from 2006 to 2018 for the entire territory of the European Union. Only the LUCAS data from 2018 were selected as the crop validation samples for this study, as Sentinel-2 L2A data used for crop mapping have been available since 2017. Due to the different classification schemes of CDL and LUCAS, we harmonized the crop types associated with corn and wheat. The crop names and codes before and after harmonization were listed in Tables 2 and 3 summarized the number of LUCAS validation samples for each mapping area. Both CDL and LUCAS data were freely accessible on the GEE platform.

Table 2. Harmonization of corn and wheat types in CDL and LUCAS.

CDL				LUCAS			
Original codes	Original names	Harmonized codes	Harmonized names	Original codes	Original names	Harmonized codes	Harmonized names
1	Corn	1	Corn	B16	Maize	1	Corn
22	Durum Wheat	20	Wheat	B11	Common wheat	20	Wheat
23	Spring Wheat			B12	Durum wheat		
24	Winter Wheat						

Table 3. The number of LUCAS validation samples for each mapping area.

	Germany	France	Romania	Poland	Spain	Italy
Corn	2389	2182	1208	1090	450	831
Wheat	2166	3864	861	1842	1753	1704
Total	4555	6046	2069	2932	2203	2535

2.2. Methods

2.2.1. Spectral band selection and spectral index calculation for satellite images

The original spectral bands common to both Landsat-8 and Sentinel-2 as mentioned in Section 2.1.2 were selected. In addition, a group of vegetation indices has been added in order to highlight the information on vegetation and crop growth processes. The specific formulae and meanings of each spectral index were given below:

$$\mathrm{NDVI} = \frac{NIR - red}{NIR + red} \tag{1}$$

NDVI (Tucker 1979) stands for the Normalized Difference Vegetation Index, which is used to quantify the greenness of vegetation and is useful for understanding vegetation density and assessing changes in plant growth.

$$\mathrm{EVI} = 2.5 \times \frac{NIR - red}{NIR + 6 \times red - 7.5 \times blue + 1} \tag{2}$$

EVI (Huete et al. 1997) represents the Enhanced Vegetation Index, which is similar to NDVI and can be used to quantify the greenness of vegetation. However, EVI was corrected for some atmospheric conditions and canopy background noise, and was more sensitive in densely vegetated areas.

$$\mathrm{GCVI} = \frac{NIR}{green} - 1 \tag{3}$$

GCVI (Gitelson et al. 2003) denotes the Green Chlorophyll Vegetation Index, which has a greater dynamic range for dense canopies (high leaf area index) than NDVI.

$$\mathrm{MSAVI} = \frac{1}{2} \times \left(2 \times NIR + 1 - ((2 \times NIR + 1)^2 - 8 \times (NIR - red))^{\frac{1}{2}}\right) \tag{4}$$

MSAVI (Qi et al. 1994) indicates the Modified Soil Adjusted Vegetation Index, and MSAVI minimizes the effect of bare soil on the Soil Adjusted Vegetation Index (SAVI). MSAVI was calculated as a ratio between red and near-infrared, and an inductive L-function was applied to minimize soil effect on the vegetation signal.

$$\mathrm{LSWI} = \frac{NIR - SWIR1}{NIR + SWIR1} \tag{5}$$

LSWI (Xiao et al. 2005) is the Land Surface Water Index, which utilizes the short-wave infrared and near-infrared regions of the electromagnetic spectrum. In the short-wave infrared, liquid water has strong light absorption; therefore, the LSWI index is more sensitive to the total amount of liquid water in vegetation and its soil background.

$$\mathrm{NDSVI} = \frac{SWIR1 - red}{SWIR1 + red} \tag{6}$$

NDSVI (Zhong, Gong, and Biging 2014) stands for the Normalized Differential Senescent Vegetation Index, which is associated with specific crop responses to water content.

$$\mathrm{NDTI} = \frac{SWIR1 - SWIR2}{SWIR1 + SWIR2} \tag{7}$$

NDTI (Zheng, Campbell, and de Beurs 2012) denotes the Normalized Difference Tillage Index, which is an indicator of residue cover.

Based on the above six spectral bands and seven vegetation indices (a total of 13 bands), time series data for each band were constructed to be used as input features of the machine learning model for crop classification. The method of constructing the time series data was described below.

2.2.2. Gap-free monthly time series satellite data construction

Crops are usually time-dynamic, that is, they exhibit multiple phenological periods, such as sowing, growing and harvesting over time. With satellite time-series observations, the spectral reflectance of crops in different phenological periods can be obtained, forming time-series spectral features, which is the most effective way to distinguish between different crops. Therefore, the construction of time series data is a key step in crop characterization.

Satellite remote sensing images are usually stored in tiles. When the study area is large, it contains images from multiple tiles, and the imaging dates and time intervals of the time series images contained in different tiles will be different. In addition, due to cloud cover and other factors, different scenes or different pixels in the same scene will have different time observation frequencies. To ensure that each pixel in the entire study area has the same temporal interval so that the training area and the mapping area have consistent time series feature representation, it is necessary to synthesize the original satellite images covering the entire study area with equal time intervals. In this study, we took month as the time interval and adopted the median compositing algorithm for the construction of monthly time series data. The specific processing is as follows:

(1) Setting temporal and spatial filters to select all available Landsat-8 and Sentinel-2 satellite images acquired in the same month covering the specific study area;
(2) Selecting the original spectral bands and calculating the selected vegetation index bands according to Section 2.2.1;
(3) Cloud masking was performed using the 'QA_PIXEL' band for Landsat-8 and the 'QA60' band for Sentinel-2;
(4) Calculating the median of all valid observations within the same month for each pixel and each band.

Compared to other time intervals (e.g. 10 days, seasons), monthly time series can dramatically reduce the amount of data as well as preserve the phenological variability of crops (Liu et al. 2023). In addition, compared to other compositing algorithms (e.g. maximum NDVI compositing, weighted score compositing), the median compositing algorithm is easy to implement, computationally efficient, and has high synthesis quality (Meng et al. 2023; Xie et al. 2019). Although the temporal compositing methods have a higher probability of obtaining a gap-free time series, gaps in the monthly time series may still occur for some regions. This is due to the lack of valid observations caused by the few image data acquired or highly frequent cloud cover. Therefore, after obtaining the monthly time series composites, the gaps of time series were filled using simple linear interpolation (Griffiths, Nendel, and Hostert 2019; Tran et al. 2022) as follows:

$$\hat{\rho}_2 = \rho_1 + (t_2 - t_1) \times \frac{\rho_3 - \rho_1}{t_3 - t_1} \tag{8}$$

where $\hat{\rho}_2$ is the interpolated spectral reflectance or vegetation index value for month t_2, and ρ_1 and ρ_3 are the valid spectral reflectance or vegetation index values for the months (t_1 and t_3) closest to t_2 before and after month t_2. Through the above process, the gap-free time series data in 13 bands for the entire growing season of the crop from March to October were obtained (i.e. 104 features in total), which were used to provide the temporal spectral characterization of the crops. The inclusion of a large number of input bands can contribute to improved classification performance, as random forest model used in this study can handle numerous input variables effectively and tend to benefit from the additional information embedded in high-dimensional feature spaces. The relevant variable importance tables were presented in the Section I of the Supplemental Material to illustrate the relative contribution of each feature used in the classification process.

2.2.3. Crop classification model development and large-area crop classification based on model transfer

The acquisition of training samples is an important prerequisite for supervised classification. The relationship between crop type and predictor variable is obtained by labeling the predictor variables with crop types from training samples. This relationship is input into a machine learning model for training to obtain a crop classification model (i.e. mapping from features to crop types). Inputting the predictor variables from a new study area into this model yields the crop classification results for that study area.

In this study, CDL data were utilized for corn and wheat training sample extraction. Although CDL had high classification accuracy and was widely used as a source of training samples, refinement of CDL data was conducted in order to further improve the quality of training samples. Because of the large potential classification errors associated with object edges, class edge pixels were eliminated from the CDL crop classification map. The class edges were defined as follows: a pixel was identified as an edge pixel if any of its eight neighboring pixels had a different land cover type from that pixel. Corn and wheat training samples were extracted based on the refined CDL data. A stratified random sampling method was utilized to independently and spatially randomly select 3000 sample points on corn and wheat types, respectively (6000 in total). A set of features consisting of monthly time series satellite observations with six spectral bands and seven vegetation indices were extracted using the selected training samples and input to the random forest (RF) algorithm to develop the crop classification model. The RF algorithm (Breiman 2001) is a non-parametric machine learning algorithm that constructs multiple decision trees by randomly selecting a subset of training samples and a subset of features. New instance is classified via the multiple decision trees, and the label of it is determined by majority vote. For the parameter setting of RF model, we specified the number of trees as 500, and all other parameters were kept at their default values. The RF algorithm was chosen due to its computational efficiency, high classification accuracy achieved through ensemble learning (Belgiu and Drăguț 2016), and its straightforward implementation within the GEE platform for large-area application (Phalke et al. 2020; Thenkabail et al. 2021; Zhao et al. 2022). In this study, the large-scale crop maps were generated using the RF model in GEE because a large amount of computation was required to obtain the classification results for each pixel. For the spatiotemporal classification experiments, a locally deployed RF model was applied to the exported training and validation sample data. Since the accuracy assessment relied solely on the classification outcomes of the validation samples, this alternative approach was more computationally efficient when conducting experiments across various scenarios.

The crop classification model established in the training region using the above method can be applied to other target regions for crop classification and mapping. In order to investigate the spatial and temporal transferability of the crop classification model over large areas, the following sets of classification experiments were carried out:

(1) Corn and wheat mapping for six European countries based on the crop classification model derived from the state of Kansas, USA, to explore the spatial transferability of crop classification models between large distant areas;
(2) Corn and wheat mapping in European countries based on crop classification models derived from different years in Kansas, USA, to explore the temporal transferability of crop classification models;
(3) Corn and wheat classification for European countries based on crop classification models for different states and years in the U.S. to explore the performance differences of classification models derived from different training regions/years, as well as the effect of regional/yearly aggregation of training samples on the transferability of the models.

2.2.4. Accuracy assessment

In order to quantitatively evaluate the classification accuracy of the transferred model, the classification results obtained using transferred model were compared with *in-situ* survey samples (i.e. LUCAS data). The quantitative accuracy metrics adopted include confusion matrix, overall accuracy (OA), producer accuracy (PA) and user accuracy (UA) (Chakhar et al. 2020; Li et al. 2023; Tran et al. 2022). OA is the ratio of the number of correctly classified samples to the total number of samples, which indicates the overall classification accuracy of the classification model on all categories. PA is the ratio of the number of samples classified into a category to the number of all reference samples in that category, reflecting the rate at

which that category is missed by the classification model. UA is the ratio of the number of samples classified into a category to the number of all reference samples classified into that category, which reflects the rate at which that category is incorrectly classified by the classification model. Both PA and UA characterize the classification performance of classifier on a single category.

In addition, the classification results were compared with those obtained using the models derived from the mapping area itself. Specifically, the LUCAS samples for any one country (i.e. mapping area) were divided into training and validation samples at a ratio of 7:3. The training samples were used to develop the RF model for mapping area, and the validation samples were used for accuracy evaluation of the two sets of classification results. This experiment aimed to explore the differences in quantitative crop classification accuracy between transferred model and local model, as well as the differences in the spatial pattern of the resulting crop classification maps.

3. Results

3.1. European country crop classification results derived using the Kansas models

Figure 2 showed the national-scale corn and wheat classification results from 2018 for Germany, France, Romania, Poland, Spain, and Italy. These results were generated using the crop classification models trained in Kansas, USA from 2019. Noted that only pixels labeled as corn or wheat in the EUCROPMAP reference dataset (d'Andrimont et al. 2021) were used for classification. Pixels corresponding to non-cropland or other crop types were excluded to ensure a clear corn/wheat crop classification task. The quantitative accuracy evaluation of the classification results for each country was shown in Table 4. The overall classification accuracies for corn and wheat in Germany, France, Romania, Poland, Spain, and Italy were 93.8%, 95.4%, 88.1%, 93.3%, 96.8%, and 95.0%, respectively. The lowest overall accuracy can be observed in Romania with 88.1%, while Spain obtained the highest overall accuracy of 96.8%. Additionally, high overall classification accuracies were also achieved in the remaining countries, all above 93%. In terms of single-category classification accuracies, corn's producer accuracies ranged from 88.3% to 91.9% and user accuracies ranged from 90.1% to 99.1%. The highest omission and commission rates for corn were found in Poland and Romania, respectively, whereas Italy had the lowest omission rate and Germany had the lowest commission rate.

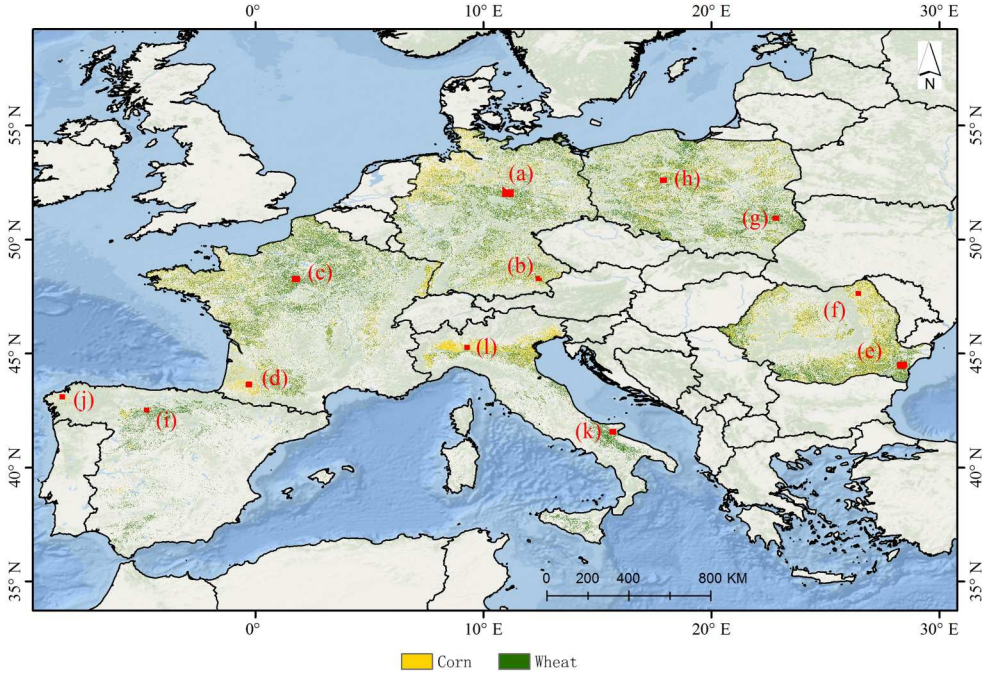


Figure 2. The corn and wheat classification results at 30 m resolution for six European countries in 2018 obtained using the models trained in Kansas, USA from 2019. (a) through (l) in red indicated the geographic locations of the subgraphs in Figure 6 showing the detailed maps.

Table 4. Confusion matrices and accuracy assessment of crop classification results for six European countries in 2018 derived using the models trained in Kansas, USA from 2019.

Germany				France			
	Corn	Wheat	PA		Corn	Wheat	PA
Corn	2124	265	88.9	Corn	1947	235	89.2
Wheat	19	2147	99.1	Wheat	41	3823	98.9
UA	99.1	89.0	93.8	UA	97.9	94.2	95.4
Romania				**Poland**			
	Corn	Wheat	PA		Corn	Wheat	PA
Corn	1080	128	89.4	Corn	962	128	88.3
Wheat	119	742	86.2	Wheat	69	1773	96.3
UA	90.1	85.3	88.1	UA	93.3	93.3	93.3
Spain				**Italy**			
	Corn	Wheat	PA		Corn	Wheat	PA
Corn	405	45	90.0	Corn	764	67	91.9
Wheat	26	1727	98.5	Wheat	60	1644	96.5
UA	94.0	97.5	96.8	UA	92.7	96.1	95.0

Notes: For each matrix, row and column represented the reference and classified data respectively. The numbers on a gray background indicated the overall accuracies. PA and UA denoted produce accuracy and user accuracy. All accuracy values were in %.

Wheat's producer accuracies ranged from 86.2% to 99.1% and user accuracies ranged from 85.3% to 97.5%. The highest omission and commission rates for wheat were also found in Romania, whereas Germany and Spain had the lowest omission and commission rates, respectively. The quantitative accuracy evaluation indicated that the application of corn and wheat classification models trained in Kansas, USA, to European countries was able to obtain highly accurate national-scale crop classification results, showing good spatial transferability.

3.2. European country crop classification based on the Kansas models from different years

Figure 3 provided the accuracy comparison of the European national-scale crop classification results from 2018 generated using the crop classification models trained in Kansas, USA from different years (2019–2023). As can be seen, the classification models from different years in Kansas achieved an overall classification accuracy of 90.0–95.0% in Germany, 93.0–95.8% in France, 85.2–88.1% in Romania, 91.0–93.0% in Poland, 95.4-96.8% in Spain, and 94.5–95.8% in Italy. Thus, the application of the classification models for different years in Kansas to European countries can yield high classification accuracy. In addition, the variations in the overall accuracy of the classification models across years were 2.0%, 1.2%,

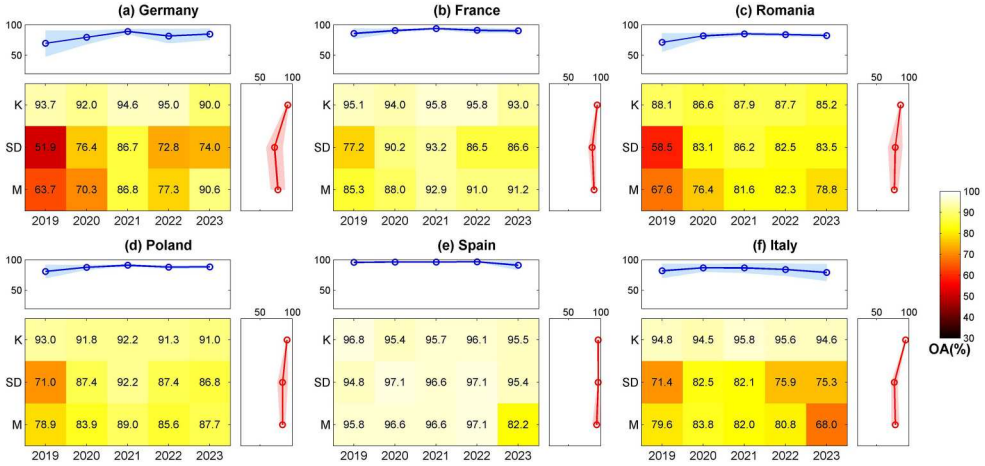


Figure 3. Accuracy evaluation and comparison of crop classification results for European countries obtained using the crop classification models trained in diverse areas (states) of USA from different years. For each subplot, the heat map represented the distribution of overall accuracy with its values superimposed on each grid of the heat map. The plots on the upper and right sides of the heat map showed the changes in the accuracies averaged over regions and years, respectively. Noted that the abbreviations of K, SD and M indicated Kansas, South Dakota, and Minnesota State.

1.2%, 0.8%, 0.6%, and 0.6% for Germany, France, Romania, Poland, Spain, and Italy, respectively. Overall, the performance of the classification models from different years changed very little for the same country. Compared to the other four years, the Kansas crop classification model from 2019 achieved the highest overall classification accuracies in Romania, Poland, and Spain; while in Germany, France and Italy, the overall classification performance of the 2021 or 2022 Kansas model was better than that of other years' classification models. The above results suggested that the corn and wheat crop classification models from different years in the same training region have similar classification performance for the same country mapped, showing good temporal transferability.

3.3. European country crop classification based on the models trained in different US states

Three crop classification models developed using the training data from three different US states (i.e. Kansas, South Dakota, and Minnesota) were applied to the six European countries for crop mapping. The quantitative accuracy evaluation of these classification results could also be found in Figure 3. Across all study areas mapped, comparisons among Kansas, South Dakota, and Minnesota classification models in terms of overall accuracy showed a largely consistent pattern, that is, the Kansas model had the highest accuracy, followed by Minnesota, and the South Dakota model performed the worst. Additionally, the performance differences of models trained in different regions were usually greater than those of models trained in the same region across different years. For example, the variations in the overall accuracy of the classification models from different states were found to be most dramatic in 2019, i.e. 21.5%, 9.0%, 15.2%, 11.1%, 1.0% and 11.9% for Germany, France, Romania, Poland, Spain and Italy, respectively. It was evident that, with the exception of Spain, the models derived from different states showed large differences in classification performance for the same country. Thus, the Kansas model had the best spatio-temporal transferability compared to the Minnesota and South Dakota models. On the other hand, although the classification performance of the Minnesota and South Dakota models decreased substantially in Germany, France, Romania, Poland and Italy, it maintained a consistently high classification accuracy in Spain showing robustness to the classification models from different training regions.

Furthermore, in order to explore the effect of the spatio-temporal aggregation of training samples on model transferability, in addition to training in individual state and year, the training samples from the states merged or years merged were also used to train a crop classification model covering a broader spatial extent or temporal span, which was then applied to the European countries. The performance of the classification models derived from the regionally/yearly aggregated samples were illustrated in Figure 4. The results of the accuracy evaluation revealed that models trained on aggregated samples, whether across

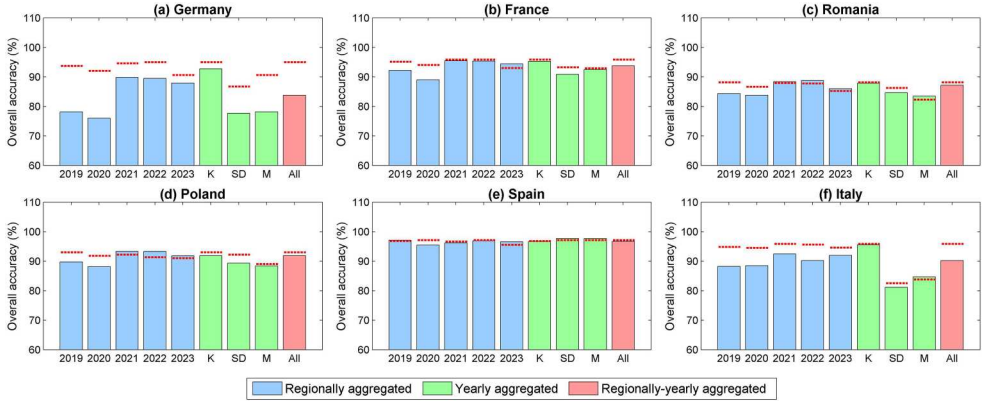


Figure 4. The overall accuracies of crop classification results for each study area derived from the models trained using the spatial-temporally aggregated samples. The blue, green and red bars represent the regionally aggregated, yearly aggregated and regionally-yearly aggregated, respectively. For example, '2019' means the merged K, SD and M in 2019 (i.e. spatial aggregation), 'K' means the merged K from 2019 to 2023 (i.e. temporal aggregation), and 'All' means the merged K, SD and M from 2019 to 2023 (i.e. spatial-temporal aggregation). The acronyms of K, SD and M have the same meanings as in Figure 3. Note that the short red dashed line above the bars indicate the optimal value if an individual year or region is used (i.e. not aggregated).

regions, years, or both, generally exhibit lower classification accuracy when applied to crop classification in Germany, France, Romania, Poland, Spain, and Italy, compared to models trained on single regionally/yearly optimal samples. Specifically, models derived from a single optimal region outperformed those trained on regionally aggregated samples by 8.9%, 1.5%, 0.8%, 0.6%, 0.2%, and 4.8% on average for Germany, France, Romania, Poland, Spain, and Italy, respectively. Similarly, models built using a single optimal year outperformed those trained on temporally aggregated samples by 7.9%, 1.1%, 0.2%, 1.6%, −0.4%, and 0.2% on average for the corresponding countries. Furthermore, models trained on fully aggregated samples across both regions and years were outperformed by those derived using single optimal region or year samples, with accuracy reductions of 11.2%, 2.0%, 1.0%, 1.2%, 0.4%, and 5.6% for Germany, France, Romania, Poland, Spain, and Italy, respectively. These findings suggest that while aggregated samples can enhance model robustness in some contexts, they may diminish the model's ability to capture region- or year-specific characteristics affected by climatic factors and potential confounders (e.g. soil type, cultivation practices) that are critical for accurate classification. It can also be explained as a shift in the merged source domain, making it less effective than the original source domain that was more similar to the target domain. All study areas show the potential sensitivity of the crop classification performance to region-specific or year-specific variations of training samples, with Germany and Italy being the most significant. It should also be noted that regional aggregation leads to a greater reduction in classification accuracy compared to temporal aggregation. This pattern suggests that regional aggregation introduces greater heterogeneity into the training dataset. In contrast, temporal aggregation likely incorporates a narrower range of variability, as intra-annual patterns, such as crop phenology, may remain relatively stable across years. The greater sensitivity to regional aggregation may also reflect the diversity of agro-ecological zones, climate types, and management practices across regions, which can create challenges for a single aggregated model to generalize effectively (Hao et al. 2020). Overall, these results emphasize the importance of considering the spatio-temporal similarity between training and mapping areas and suggest that targeted regional models or region-specific sampling strategies may outperform models based on broader regional aggregations, as overgeneralized sample aggregation may compromise model performance in geographically diverse contexts.

3.4. Comparison between transferred and local models

All the previous classification results for the European countries were obtained based on the classification models from the US regions without using any training samples of the areas being mapped. Despite the high crop classification accuracies achieved, comparisons with the classification results of the local models were needed to explore whether the transferred models differ from the local models? How big was the difference? It was worth noting that the local model here referred to the classification model obtained by using the crop training samples from the mapped area itself. In a broader sense, local and transferred models belong to the target-domain and source-domain models, respectively.

To obtain classification results from the local model and to make a fair quantitative comparison with the results of the transferred model, we split the LUCAS sample data into training and validation sets in a 7:3 ratio. 70% of the LUCAS samples were used to build the local model with exactly the same input features as the earlier transferred model, and the remaining 30% were used to validate the classification results of the local and transferred models. Figure 5 illustrated the quantitative accuracy comparison between local and transferred model classification results, where the transferred model results were obtained based on the optimal classification models from previous US states. The differences between the overall classification accuracies of transferred and local models were −2.0%, −1.4%, −1.3%, −0.5%, −0.6%, and −0.8% for Germany, France, Romania, Poland, Spain, and Italy, respectively. For all study areas, the overall accuracy of transferred model was slightly lower than that of local model, with decreases ranging from 0.5% (Poland) to 2.0% (Germany). This observation aligned with the findings of previous research that reported the similar patterns (Hao et al. 2020). Overall, the transferred models were almost as accurate as the local models for crop classification.

In addition to the quantitative accuracy evaluation, crop classification maps derived using the transferred model and the local models were compared in some close-up areas to examine the similarities and differences in the spatial distribution of crop classification results. Two regions in each country were selected (see Figure 2 for the geographic location of each selected region) for the comparison of the spatially detailed

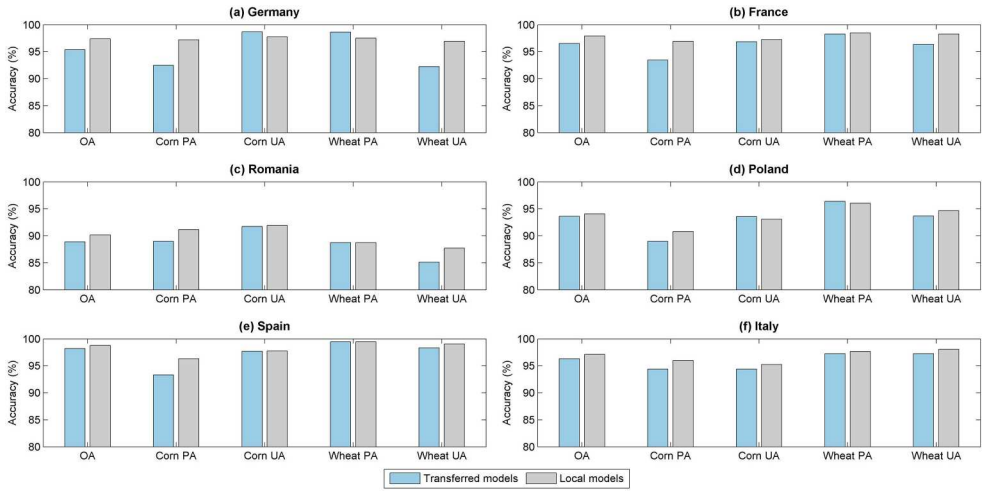


Figure 5. Comparative crop classification accuracy between transferred and local models for each study area.

maps, as illustrated in Figure 6. Evidently, there was a very high degree of agreement in the spatial distribution of corn and wheat in the classification results of transferred and local model. From the above quantitative and qualitative evaluations, the overall classification accuracies of the transferred model and the

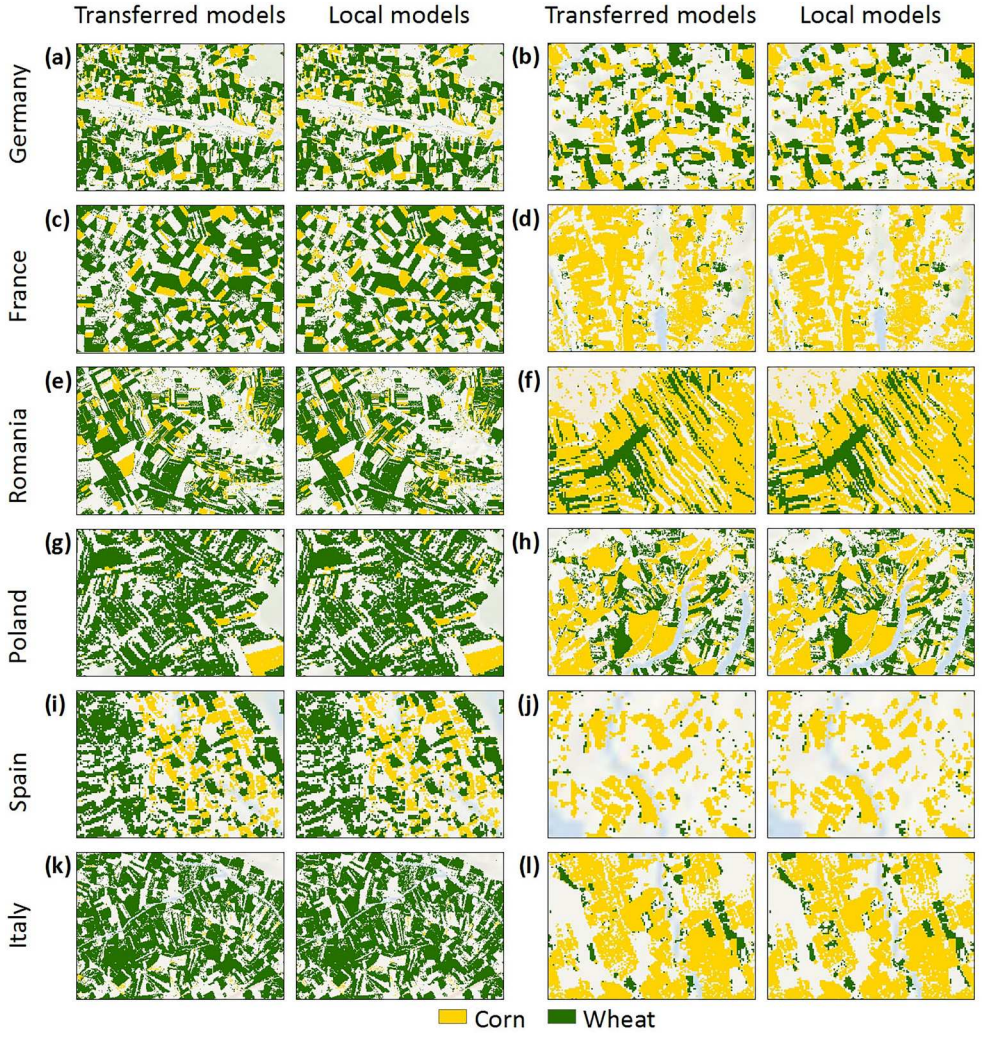


Figure 6. The zoomed-in crop classification results at 30 m resolution from the transferred model and the local model, and the geographic locations of each subregion from (a) to (l) were indicated in Figure 2.

local model did not differ much, and the resulting corn and wheat classification maps were highly consistent in their spatial distribution. The transferred models developed using training samples from external regions have comparable performance with the local models which use the training samples from the mapped areas itself for the national-scale crop classification.

3.5. Effect of climatic factor on spatio-temporal transferability of crop classification models

From the results previously obtained, the crop classification performance of the models trained in different areas and years varied considerably. Why the difference is so big and what drives it? In this section, we investigated the effect of climatic factors on the various classification models derived from different training regions and time periods. Crop phenology is often influenced by meteorological conditions that alter its spectral characteristics (Gao and Zhang 2021), and differences in meteorological factors across regions or years may lead to degradation of the model's generalization ability. We selected two climatic variables, temperature and precipitation, and used them to measure the similarity between regions during different years. Specifically, the monthly time series temperature (precipitation) data were extracted from training and mapping areas, and the Euclidean distance between them was calculated to characterize the similarity across regions, i.e. the smaller the Euclidean distance, the more similar the regions. Then, Pearson Correlation Coefficients (PCC) (Berman 2016) were used to explore the relationship between overall classification accuracy and regional similarity indicated by temperature (precipitation). PCC is a statistical measure of linear correlation between two variables with values ranging from −1 to 1. It reveals the magnitude and direction of the correlation between variables. A value closer to 1 indicates a positive correlation between the two variables, i.e. when one increases, the other increases accordingly. And vice versa. Figure 7 illustrated the correlation coefficients between overall classification accuracy and regional similarity as measured by temperature (precipitation) pattern. Results suggested that for all study areas except Spain, correlation coefficients between overall classification accuracy and temperature pattern similarity were greater than 0.6 with strong statistical significance. The sample size for each group of the correlation analysis is 15 (i.e. 3 training areas × 5 training years). Although the sample size is relatively small, the relationship between temperature similarity and accuracy remains strong statistical significance. Additionally, we combined all the 'temperature similarity-accuracy' sample pairs from all six groups (i.e. a sample size of $15 \times 6 = 90$) for an integrated correlation analysis, yielding an overall positive correlation coefficient (0.36) with strong statistical significance ($p < 0.01$). Therefore, the notably low correlation coefficient in Spain may be due to anomalous outliers caused by the small sample size, but it does not affect the overall correlation and statistical significance of the 'temperature similarity-accuracy' relationship. Unlike the situation with temperature, small and statistically insignificant correlation coefficients were observed for the relationship between overall classification accuracy and precipitation pattern similarity. This may reflect irrigation-dependent agriculture in the U.S./EU, decoupling natural precipitation from model performance. Thus, the performance of the transferred model was strongly and positively correlated with the similarity of temperature patterns between training and mapping regions.

In addition to correlation analysis, regression analysis (Kotu and Deshpande 2019) was applied to quantitatively describe the linear relationship between the overall classification accuracy and temperature pattern similarity. The regression analysis results were provided in Figure 8. For each subplot, the *x*-axis represented

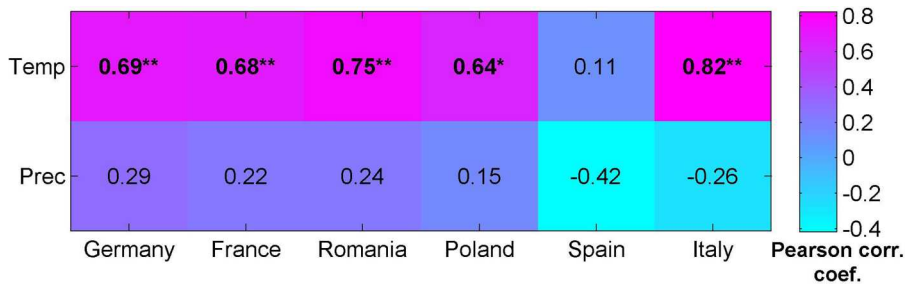


Figure 7. Pearson correlation coefficients between overall classification accuracy and region similarity as measured by temperature (precipitation) pattern for each country. Noted that ** and * denoted $p < 0.01$ and $p < 0.05$, respectively.

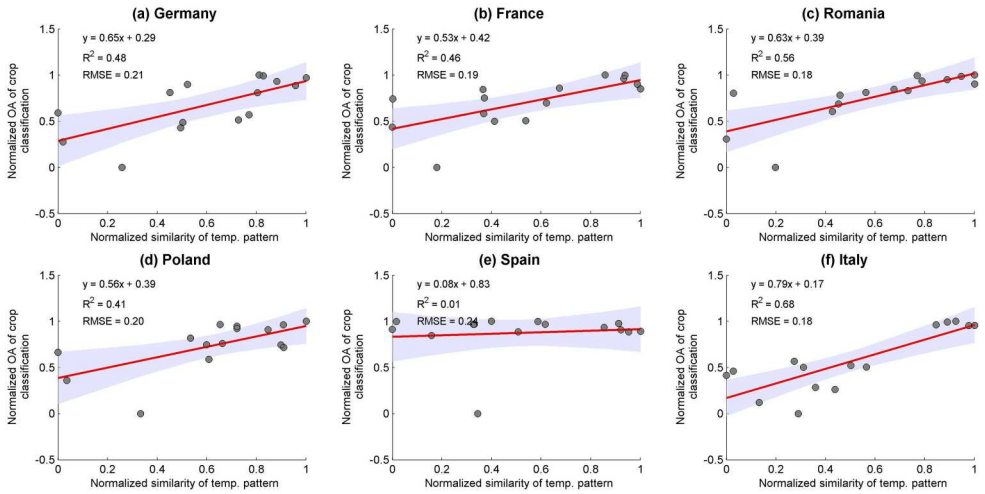


Figure 8. Linear regression analysis for the relationship between the overall classification accuracy of transferred models and the temperature pattern similarity between training and mapping areas.

the temperature pattern similarity between training area (year) and that country, and *y*-axis represented the overall accuracies obtained using the crop classification models derived from that training area (year). As indicated in Figure 8, strong linear relationships (with R^2 ranging from 0.41–0.68 and RMSE between 0.18 and 0.21) were obtained for five countries (excluding Spain), i.e. the variation of OA could be explained by the variation of temperature pattern similarity. Therefore, the more similar the temperature patterns between the training and mapping areas, the better the classification performance of the model derived from that training area. This finding was consistent with previous study (Wang, Azzari, and Lobell 2019) which reported that high accuracy will be achieved in neighboring regions with similar growing degree days (GDD, the function of temperature).

4. Discussion

4.1. Advantages of transferred models

Multi-type crop information is usually obtained through field surveys, and when it comes to large-area crop type mapping, a huge number of ground samples need to be collected as training data, which is very time-consuming and labor-intensive leading to high data collection costs (Fowler, Waldner, and Hochman 2020; Lei et al. 2024; Li et al. 2023). As a result, crop classification maps for large regions (e.g. national scales) are difficult to obtain, limited by the availability of training samples. Currently, the U.S. has created a national-scale, long time-series, high-accuracy crop type map. The crop classification machine learning model could be developed using the crop labeling data and satellite remotely sensed images of the U.S., and applied to other regions with absence of ground samples to address the lack of crop type maps over there. To explore the spatio-temporal transferability of the U.S. crop classification model at national scale, this study utilized the U.S. CDL crop type product and Landsat-8/Sentinel-2 satellite imagery to build a series of machine learning models which were then applied to the Landsat-8/Sentinel-2 satellite imagery of European countries for national-scale crop classification. Despite the fact that the U.S. and European countries are located on different continents and have different climatic types and ecological environment, the U.S. crop classification models show great spatial and temporal transferability (depending on the choice of training region), and are able to obtain classification accuracies comparable to those of local models and without the need to collect any training samples locally, thus providing a viable technical approach for crop type mapping over large areas with few or no ground samples.

4.2. Importance of inter-regional similarity in annual temperature patterns for transferred models

As we found in this study the significant performance differences in crop classification models when transferred across different regions (years), particularly under diverse climatic contexts. Through quantitative

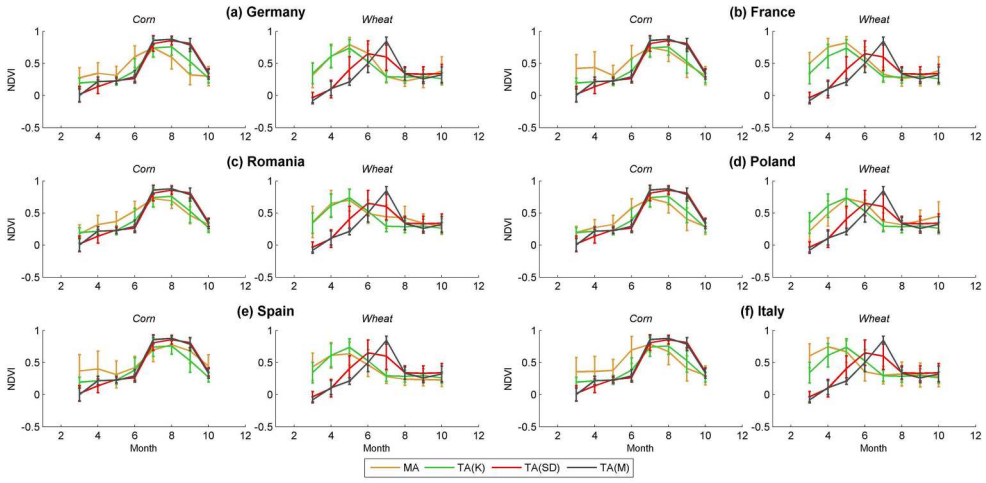


Figure 9. The spectral reflectance curves of crops from the three training areas (i.e. Kansas, South Dakota and Minnesota) and six mapping areas (i.e. Germany, France, Romania, Poland, Spain and Italy). In legend, the MA and TA denote mapping and training areas, respectively. K, SD and M have the same meanings as in Figure 3.

attribution analysis, we identified that the similarity of temperature patterns between the training and mapping regions played a crucial role in determining the classification accuracy of the models. Specifically, when the temperature profiles of the training and mapping regions were more similar, the classification performance of the models improved significantly. This finding suggests that temperature, as a key climatic factor influencing crop growth and distribution, may alter farming practices, crop phenological patterns and seasonal characteristics, which in turn affects the remotely sensed reflectance signatures used by classification models. To further validate this finding, we compared the spectral reflectance curves of crops from different regions (Figure 9) alongside their respective intra-annual temperature time series (Figure 10). The analysis revealed that regions with similar temperature profiles exhibited comparable crop spectral curves. For example, Kansas exhibits the most similar temperature profiles to each of the mapping areas, and the crop spectral curves in Kansas also closely match those in each mapping area, especially for wheat. As shown in Figures 9 and 10, Kansas and European countries experience relatively mild winter temperatures, and the peak of the NDVI curve for wheat in these areas occurs around May, suggesting the dominance of winter wheat cultivation in Kansas and European countries (USDA-NASS n.d.– a; Eurostat n.d.). In

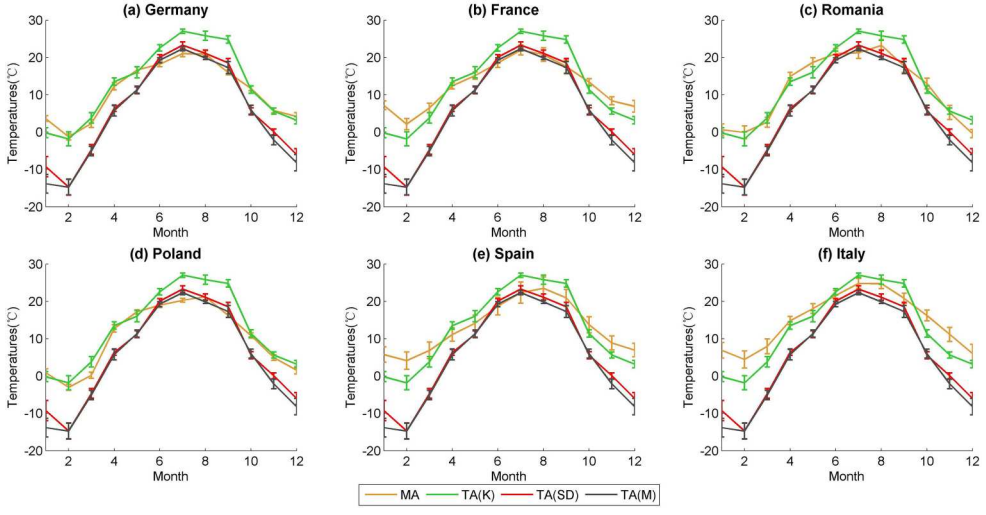


Figure 10. The intra-annual temperature profiles for the three training areas (i.e. Kansas, South Dakota and Minnesota) and six mapping areas (i.e. Germany, France, Romania, Poland, Spain and Italy). The acronyms in legend have the same meanings as in Figure 3.

contrast, Minnesota, located at the highest latitude among the studied regions, has cold winters, and its wheat NDVI peak occurs around July. This implies that spring wheat is primarily grown in Minnesota (USDA-NASS n.d. – b). South Dakota, situated between Kansas and Minnesota in latitude, experiences winter temperatures slightly higher than Minnesota but lower than Kansas. The peak NDVI month of wheat in South Dakota falls between those of Kansas and Minnesota, indicating the presence of both winter and spring wheat cultivation in this region (USDA-NASS n.d. – c). This observation aligns with the previously obtained results, confirming that the model trained on Kansas data performs the best in terms of classification accuracy compared to those trained on the other two states. This suggests that temperature-driven phenological synchronization plays a significant role in improving model transferability and classification accuracy across different regions. Consequently, the temperature pattern similarity between regions highlights the importance of considering climatic differences when applying models across different areas, and is helpful for selecting the optimal training areas to build crop classification models with better transferability. Specifically, temperature pattern similarity between regions, indicated by the Euclidean distance of intra-year temperature profiles, can be used as an indicator for source-region selection when applying pre-trained crop classification models to new areas. In situations where multiple source regions are available, prioritizing those with higher temperature similarity to the target region can help minimize domain shifts, and maintain higher classification performance before further possible tuning of the model.

4.3. Mechanistic explanation of temperature factors and potential effects of non-temperature factors

For large-scale crop mapping, our finding that higher temperature similarity between the source and target regions improves transfer accuracy can be understood through the theoretical lens of phenological synchrony in crop growth. From an agroecological perspective, temperature regimes are primary drivers of crop development rates and growth stage timing (Estrella, Sparks, and Menzel 2007; Luo, Bange, and Clancy 2014; Oteros et al. 2015). When two regions share similar thermal conditions, their crops tend to follow comparable phenological calendars, leading to temporal alignment of key phenological stages. From a remote sensing and domain adaptation standpoint, this synchrony produces spectrally homologous time-series trajectories in vegetation indices and other spectral metrics. Such homology minimizes the domain shift between training and application datasets, thereby enhancing the generalization capacity of the classification model. Conversely, larger temperature divergences disrupt this alignment, introducing temporal and spectral mismatches that reduce model transferability.

Beyond temperature effects, non-temperature factors – including sowing dates, cultivar differences, soil properties, and management practices – may influence crop phenology (Gao and Zhang 2021; He, Jin, and Yu 2020; Rezaei, Siebert, and Ewert 2017). For example, management practices such as changing sowing dates or using cultivars with longer growing periods (He, Jin, and Yu 2020; Rezaei, Siebert, and Ewert 2017) can shift the timing of key phenological stages. In regions where thermal regimes are relatively uniform but agronomic conditions are heterogeneous, these non-temperature factors may exert a stronger influence than climatic similarity, as they directly shape phenology and spectral dynamics, thereby affecting model generalization across regions. However, due to the limited availability of data on these non-temperature factors, we are currently unable to assess their impact on model transferability. Compared to other factors, meteorological data, such as temperature and precipitation, is readily available and measurable particularly for large areas. We identify annual temperature similarity as a statistically significant and practically useful indicator among the tested environmental variables. Future work will consider assessing more refined influence factors and developing a more robust and transferable crop classification model.

4.4. Limitations

This study has the following limitations. First, the classification focused on major food crops (corn and wheat), chosen for their large planting areas common to both U.S. training regions and European mapping regions. Future work will extend the transferred classification model to more crop types. Second, accuracy validation relied on the high-quality, publicly available LUCAS survey data, which restricted experiments to

Europe. Expanding to additional countries across different continents will allow the testing of transferred model in more diverse geographic environments. Last, although this study focused on RF, we also conducted some tests on deep learning (DL). Preliminary results and analyses reveal varying transferability and sample aggregation effects depending on model types and regions (see Section II and III of the Supplemental Material). Given that an extensive cross-model comparative assessment is beyond the current scope, a systematic exploration of DL models for large-scale crop classification will be addressed in future work.

5. Conclusion

In this study, a national-scale crop type mapping method based on the transfer of classification models was developed. We further explored the spatio-temporal transferability of classification models at the national scale as well as the differences in the transferability of classification models obtained from different training areas and years. The main findings of this study were as follows: (i) Classification model obtained from Kansas, 2019 can achieve high-accuracy crop classification in European countries, with an overall accuracy of 88.1–96.8% showing good spatial transferability. Additionally, crop classification models obtained from different years in Kansas, applied to European countries presented very little difference (ranging from 0.6% to 2.1%) in overall classification accuracy showing good temporal transferability. (ii) The performance of classification models from different training regions in the US applied to European countries varied widely (for example, overall accuracy differences between 1.0% and 21.6% from 2019), with Kansas model having the highest accuracy, Minnesota the second highest, and South Dakota the lowest; in addition, the classification models from the training areas or years merged performed less well on average than those from optimal individual training area or year, indicating that the expansion of the training area or temporal span does not necessarily result in a model with better transferable behavior. (iii) The transferred models have comparable crop classification performance (overall accuracy difference less than 2.0%) to the local models derived using training samples from mapping area itself. (iv) Temperature similarity between regions significantly affects model transferability. Specifically, the more similar the temperature patterns are, the better the performance of the transferred crop classification models. This study proposed a feasible approach for crop type mapping of large areas without prior sample data and offered valuable insights for the selection of optimal training areas to develop the transferred model.

Acknowledgements

The authors thank the U.S. Geological Survey (USGS) for Landsat data, the European Space Agency (ESA) for Sentinel-2 data, the U.S. Department of Agriculture (USDA) for CDL crop type data, the European Commission Joint Research Centre (JRC) for LUCAS in-situ land cover and use database, and the European Centre for Medium-Range Weather Forecasts (ECMWF) for climate reanalysis dataset.

Author contributions

Shuai Xie: Conception and design. **Shuai Xie and Naijie Zhang:** Acquisition, analysis and interpretation of the data; drafting of the paper. **Liangyun Liu and Lin Sun:** Revising it critically for intellectual content. **Shuai Xie, Naijie Zhang, Liangyun Liu and Lin Sun:** Final approval of the version to be published. All authors agree to be accountable for all aspects of the work.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by the National Natural Science Foundation of China [grant number 42301534]; the Shandong Provincial Natural Science Foundation [grant number ZR2023QD072]; and the Youth Innovation Team Support Plan of Shandong Provincial Higher Education Institutions [grant number 2024KJN025].

Data availability statement

All datasets used in this study are freely accessible from the Google Earth Engine (https://code.earthengine.google.com/). The GEE identifiers for each asset are as follows: Air temperature and precipitation data: ee.ImageCollection ("ECMWF/ERA5_LAND/MONTHLY_AGGR"); Landsat-8 SR data: ee.ImageCollection("LANDSAT/LC08/C02/T1_L2"); Sentinel-2 SR data: ee.ImageCollection("COPERNICUS/S2_SR_HARMONIZED"); LUCAS validation samples: ee.FeatureCollection("JRC/LUCAS_HARMO/THLOC/V1"); CDL data: ee.ImageCollection("USDA/NASS/CDL"). The training and validation data used in this study is publicly available via https://doi.org/10.6084/m9.figshare.29925809.v1.

References

Antonijević, Ognjen, Slobodan Jelić, Branislav Bajat, and Milan Kilibarda. 2023. "Transfer Learning Approach Based on Satellite Image Time Series for the Crop Classification Problem." *Journal of Big Data* 10 (1): 54. https://doi.org/10.1186/s40537-023-00735-2.

Belgiu, Mariana, and Lucian Drăguț. 2016. "Random Forest in Remote Sensing: A Review of Applications and Future Directions." *ISPRS Journal of Photogrammetry and Remote Sensing* 114:24–31. https://doi.org/10.1016/j.isprsjprs.2016.01.011.

Berman, Jules J. 2016. "Chapter 4 – Understanding Your Data." In *Data Simplification*, edited by Jules J. Berman, 135–187. Boston: Morgan Kaufmann.

Boryan, Claire, Zhengwei Yang, and Liping Di. 2012. "Deriving 2011 Cultivated Land Cover Data Sets Using Usda National Agricultural Statistics Service Historic Cropland Data Layers." Paper Presented at the 2012 IEEE International Geoscience and Remote Sensing Symposium.

Boryan, Claire, Zhengwei Yang, Rick Mueller, and Mike Craig. 2011. "Monitoring US Agriculture: The US Department of Agriculture, National Agricultural Statistics Service, Cropland Data Layer Program." *Geocarto International* 26 (5): 341–358. https://doi.org/10.1080/10106049.2011.562309.

Breiman, Leo. 2001. "Random Forests." *Machine Learning* 45 (1): 5–32. https://doi.org/10.1023/A:1010933404324.

Chakhar, Amal, Damián Ortega-Terol, David Hernández-López, Rocío Ballesteros, José F. Ortega, and Miguel A. Moreno. 2020. "Assessing the Accuracy of Multiple Classification Algorithms for Crop Classification Using Landsat-8 and Sentinel-2 Data." *Remote Sensing* 12 (11): 1735. https://doi.org/10.3390/rs12111735.

Chastain, Robert, Ian Housman, Joshua Goldstein, Mark Finco, and Karis Tenneson. 2019. "Empirical Cross Sensor Comparison of Sentinel-2A and 2B MSI, Landsat-8 OLI, and Landsat-7 ETM+ top of Atmosphere Spectral Characteristics over the Conterminous United States." *Remote Sensing of Environment* 221:274–285. https://doi.org/10.1016/j.rse.2018.11.012.

Crawford, Christopher J., David P. Roy, Saeed Arab, Christopher Barnes, Eric Vermote, Glynn Hulley, Aaron Gerace, Mike Choate, Christopher Engebretson, and Esad Micijevic. 2023. "The 50-Year Landsat Collection 2 Archive." *Science of Remote Sensing* 8:100103. https://doi.org/10.1016/j.srs.2023.100103.

d'Andrimont, Raphaël, Astrid Verhegghen, Guido Lemoine, Pieter Kempeneers, Michele Meroni, and Marijn Van Der Velde. 2021. "From Parcel to Continental Scale – a First European Crop Type Map Based on Sentinel-1 and LUCAS Copernicus in-situ Observations." *Remote Sensing of Environment* 266:112708. https://doi.org/10.1016/j.rse.2021.112708.

d'Andrimont, Raphaël, Momchil Yordanov, Laura Martinez-Sanchez, Beatrice Eiselt, Alessandra Palmieri, Paolo Dominici, Javier Gallego, Hannes Isaak Reuter, Christian Joebges, and Guido Lemoine. 2020. "Harmonised LUCAS In-Situ Land Cover and Use Database for Field Surveys from 2006 to 2018 in the European Union." *Scientific Data* 7 (1): 352. https://doi.org/10.1038/s41597-020-00675-z.

Estrella, Nicole, Tim H. Sparks, and Annette Menzel. 2007. "Trends and Temperature Response in the Phenology of Crops in Germany." *Global Change Biology* 13 (8): 1737–1747. https://doi.org/10.1111/j.1365-2486.2007.01374.x.

Eurostat. n.d. "Crop production in EU standard humidity by NUTS 2 region." Accessed August 15, 2024. https://doi.org/10.2908/APRO_CPSHR.

Fowler, Jared, François Waldner, and Zvi Hochman. 2020. "All Pixels Are Useful, but Some Are More Useful: Efficient in Situ Data Collection for Crop-Type Mapping Using Sequential Exploration Methods." *International Journal of Applied Earth Observation and Geoinformation* 91:102114. https://doi.org/10.1016/j.jag.2020.102114.

Gao, Feng, and Xiaoyang Zhang. 2021. "Mapping Crop Phenology in near Real-Time Using Satellite Remote Sensing: Challenges and Opportunities." *Journal of Remote Sensing* 2021:8379391.

Ge, Shuang, Jinshui Zhang, Yaozhong Pan, Zhi Yang, and Shuang Zhu. 2021. "Transferable Deep Learning Model Based on the Phenological Matching Principle for Mapping Crop Extent." *International Journal of Applied Earth Observation and Geoinformation* 102:102451. https://doi.org/10.1016/j.jag.2021.102451.

Gitelson, Anatoly A., Andrés Viña, Timothy J. Arkebauer, Donald C. Rundquist, Galina Keydan, and Bryan Leavitt. 2003. "Remote Estimation of Leaf Area Index and Green Leaf Biomass in Maize Canopies." *Geophysical Research Letters* 30 (5): 1248.

Griffiths, Patrick, Claas Nendel, and Patrick Hostert. 2019. "Intra-annual Reflectance Composites from Sentinel-2 and Landsat for National-Scale Crop and Land Cover Mapping." *Remote Sensing of Environment* 220: 135–151. https://doi.org/10.1016/j.rse.2018.10.031.

Hao, Pengyu, Liping Di, Chen Zhang, and Liying Guo. 2020. "Transfer Learning for Crop Classification with Cropland Data Layer Data (CDL) as Training Samples." *Science of the Total Environment* 733:138869. https://doi.org/10.1016/j.scitotenv.2020.138869.

He, Liang, Ning Jin, and Qiang Yu. 2020. "Impacts of Climate Change and Crop Management Practices on Soybean Phenology Changes in China." *Science of the Total Environment* 707:135638. https://doi.org/10.1016/j.scitotenv.2019.135638.

Huete, A. R., H. Q. Liu, K. V. Batchily, and W. J. D. A. Van Leeuwen. 1997. "A Comparison of Vegetation Indices over a Global set of TM Images for EOS-MODIS." *Remote Sensing of Environment* 59 (3): 440–451. https://doi.org/10.1016/S0034-4257(96)00112-5.

Inglada, Jordi, Arthur Vincent, Marcela Arias, Benjamin Tardy, David Morin, and Isabel Rodes. 2017. "Operational High Resolution Land Cover map Production at the Country Scale Using Satellite Image Time Series." *Remote Sensing* 9 (1): 95. https://doi.org/10.3390/rs9010095.

Kotu, Vijay, and Bala Deshpande. 2019. "Chapter 5 – Regression Methods." In *Data Science*, edited by Vijay Kotu and Bala Deshpande, 2nd ed., 165–197. Burlington, MA: Morgan Kaufmann.

Lei, Lei, Xinyu Wang, Liangpei Zhang, Xin Hu, and Yanfei Zhong. 2024. "CROPUP: Historical Products Are All You Need? An End-to-End Cross-Year Crop Map Updating Framework without the Need for in Situ Samples." *Remote Sensing of Environment* 315:114430. https://doi.org/10.1016/j.rse.2024.114430.

Li, Haijun, Xiao-Peng Song, Matthew C. Hansen, Inbal Becker-Reshef, Bernard Adusei, Jeffrey Pickering, Li Wang, Lei Wang, Zhengyang Lin, and Viviana Zalles. 2023. "Development of a 10-m Resolution Maize and Soybean Map over China: Matching Satellite-Based Crop Classification with Sample-Based Area Estimation." *Remote Sensing of Environment* 294:113623. https://doi.org/10.1016/j.rse.2023.113623.

Liu, Xiaomi, Shuai Xie, Jiangning Yang, Lin Sun, Liangyun Liu, Qing Zhang, and Chenghai Yang. 2023. "Comparisons between Temporal Statistical Metrics, Time Series Stacks and Phenological Features Derived from NASA Harmonized Landsat Sentinel-2 Data for Crop Type Mapping." *Computers and Electronics in Agriculture* 211:108015. https://doi.org/10.1016/j.compag.2023.108015.

Luo, Qunying, Michael Bange, and Loretta Clancy. 2014. "Cotton Crop Phenology in a New Temperature Regime." *Ecological Modelling* 285:22–29. https://doi.org/10.1016/j.ecolmodel.2014.04.018.

Luo, Yuchuan, Zhao Zhang, Liangliang Zhang, Jichong Han, Juan Cao, and Jing Zhang. 2022. "Developing High-Resolution Crop Maps for Major Crops in the European Union Based on Transductive Transfer Learning and Limited Ground Data." *Remote Sensing* 14 (8): 1809. https://doi.org/10.3390/rs14081809.

Main-Knorn, Magdalena, Bringfried Pflug, Jerome Louis, Vincent Debaecker, Uwe Müller-Wilm, and Ferran Gascon. 2017. "Sen2Cor for Sentinel-2." Paper Presented at the Image and Signal Processing for Remote Sensing XXIII.

Meng, Xichen, Shuai Xie, Lin Sun, Liangyun Liu, and Yilong Han. 2023. "Evaluation of Temporal Compositing Algorithms for Annual Land Cover Classification Using Landsat Time Series Data." *International Journal of Digital Earth* 16 (1): 2574–2598. https://doi.org/10.1080/17538947.2023.2230958.

Muñoz Sabater, J. 2019. "ERA5-Land Monthly Averaged Data from 1950 to Present." *Copernicus Climate Change Service (C3S) Climate Data Store (CDS)*. Accessed August 15, 2024. https://doi.org/10.24381/cds.68d2bb30.

Orynbaikyzy, Aiym, Ursula Gessner, and Christopher Conrad. 2022. "Spatial Transferability of Random Forest Models for Crop Type Classification Using Sentinel-1 and Sentinel-2." *Remote Sensing* 14 (6): 1493. https://doi.org/10.3390/rs14061493.

Oteros, Jose, Herminia García-Mozo, Roser Botey, Antonio Mestre, and Carmen Galán. 2015. "Variations in Cereal Crop Phenology in Spain over the Last Twenty-Six Years (1986–2012)." *Climatic Change* 130 (4): 545–558. https://doi.org/10.1007/s10584-015-1363-9.

Phalke, Aparna R., Mutlu Özdoğan, Prasad S. Thenkabail, Tyler Erickson, Noel Gorelick, Kamini Yadav, and Russell G. Congalton. 2020. "Mapping Croplands of Europe, Middle East, Russia, and Central Asia Using Landsat, Random Forest, and Google Earth Engine." *ISPRS Journal of Photogrammetry and Remote Sensing* 167:104–122. https://doi.org/10.1016/j.isprsjprs.2020.06.022.

Preidl, Sebastian, Maximilian Lange, and Daniel Doktor. 2020. "Introducing APiC for Regionalised Land Cover Mapping on the National Scale Using Sentinel-2A Imagery." *Remote Sensing of Environment* 240:111673. https://doi.org/10.1016/j.rse.2020.111673.

Qadir, Abdul, Sergii Skakun, Nataliia Kussul, Andrii Shelestov, and Inbal Becker-Reshef. 2024. "A Generalized Model for Mapping Sunflower Areas Using Sentinel-1 SAR Data." *Remote Sensing of Environment* 306:114132. https://doi.org/10.1016/j.rse.2024.114132.

Qi, Jiaguo, Abdelghani Chehbouni, Alfredo R. Huete, Yann H. Kerr, and Soroosh Sorooshian. 1994. "A Modified Soil Adjusted Vegetation Index." *Remote Sensing of Environment* 48 (2):119–126. https://doi.org/10.1016/0034-4257(94)90134-1.

Rezaei, Ehsan Eyshi, Stefan Siebert, and Frank Ewert. 2017. "Climate and Management Interaction Cause Diverse Crop Phenology Trends." *Agricultural and Forest Meteorology* 233:55–70. https://doi.org/10.1016/j.agrformet.2016.11.003.

Thenkabail, Prasad S., Pardhasaradhi G. Teluguntla, Jun Xiong, Adam Oliphant, Russell G. Congalton, Mutlu Ozdogan, Murali Krishna Gumma, James C. Tilton, Chandra Giri, and Cristina Milesi. 2021. "Global Cropland-Extent Product at 30-m Resolution (GCEP30) Derived from Landsat Satellite Time-Series Data for the Year 2015 Using Multiple Machine-Learning Algorithms on Google Earth Engine Cloud." *U.S. Geological Survey Professional Paper* 1868:63.

Tran, Khuong H., Hankui K. Zhang, John T. McMaine, Xiaoyang Zhang, and Dong Luo. 2022. "10 m Crop Type Mapping Using Sentinel-2 Reflectance and 30 m Cropland Data Layer Product." *International Journal of Applied Earth Observation and Geoinformation* 107:102692. https://doi.org/10.1016/j.jag.2022.102692.

Tucker, Compton J. 1979. "Red and Photographic Infrared Linear Combinations for Monitoring Vegetation." *Remote Sensing of Environment* 8 (2): 127–150. https://doi.org/10.1016/0034-4257(79)90013-0.

USDA-NASS. n.d. – a. "2023 Kansas Agricultural Overview." United States Department of Agriculture (USDA)," National Agricultural Statistics Service (NASS). Accessed February 4, 2025. https://www.nass.usda.gov/Quick_Stats/Ag_Overview/stateOverview.php?state=KANSAS.

USDA-NASS. n.d. – b. "2023 Minnesota Agricultural Overview." United States Department of Agriculture (USDA), National Agricultural Statistics Service (NASS). Accessed February 4, 2025. https://www.nass.usda.gov/Quick_Stats/Ag_Overview/stateOverview.php?state=MINNESOTA.

USDA-NASS. n.d. – c. "2023 South Dakota Agricultural Overview." United States Department of Agriculture (USDA), National Agricultural Statistics Service (NASS)." Accessed February 4, 2025. https://www.nass.usda.gov/Quick_Stats/Ag_Overview/stateOverview.php?state=SOUTH%20DAKOTA.

USDA-NASS. n.d. – d. "Data Visualization of Commodity." United States Department of Agriculture (USDA)," National Agricultural Statistics Service (NASS). Accessed August 15, 2024. https://www.nass.usda.gov/Data_Visualization/Commodity/index.php.

Vermote, Eric, Chris Justice, Martin Claverie, and Belen Franch. 2016. "Preliminary Analysis of the Performance of the Landsat 8/OLI Land Surface Reflectance Product." *Remote Sensing of Environment* 185:46–56. https://doi.org/10.1016/j.rse.2016.04.008.

Wang, Sherrie, George Azzari, and David B. Lobell. 2019. "Crop Type Mapping Without Field-Level Labels: Random Forest Transfer and Unsupervised Clustering Techniques." *Remote Sensing of Environment* 222:303–317. https://doi.org/10.1016/j.rse.2018.12.026.

Wang, Yumiao, Luwei Feng, Zhou Zhang, and Feng Tian. 2023. "An Unsupervised Domain Adaptation Deep Learning Method for Spatial and Temporal Transferable Crop Type Mapping Using Sentinel-2 Imagery." *ISPRS Journal of Photogrammetry and Remote Sensing* 199:102–117. https://doi.org/10.1016/j.isprsjprs.2023.04.002.

Wardlow, Brian D., and Stephen L. Egbert. 2008. "Large-Area Crop Mapping Using Time-Series MODIS 250 m NDVI Data: An Assessment for the US Central Great Plains." *Remote Sensing of Environment* 112 (3): 1096–1116. https://doi.org/10.1016/j.rse.2007.07.019.

Xiao, Xiangming, Stephen Boles, Jiyuan Liu, Dafang Zhuang, Steve Frolking, Changsheng Li, William Salas, and Berrien Moore III. 2005. "Mapping Paddy Rice Agriculture in Southern China Using Multi-temporal MODIS Images." *Remote Sensing of Environment* 95 (4): 480–492. https://doi.org/10.1016/j.rse.2004.12.009.

Xie, Shuai, Liangyun Liu, Xiao Zhang, Jiangning Yang, Xidong Chen, and Yuan Gao. 2019. "Automatic Land-Cover Mapping Using Landsat Time-Series Data Based on Google Earth Engine." *Remote Sensing* 11 (24): 3023. https://doi.org/10.3390/rs11243023.

Xie, Shuai, Lin Sun, Liangyun Liu, and Xiaomi Liu. 2022. "Global Cross-Sensor Transformation Functions for Landsat-8 and Sentinel-2 Top of Atmosphere and Surface Reflectance Products within Google Earth Engine." *IEEE Transactions on Geoscience and Remote Sensing* 60:1–9.

Xu, Jinfan, Yue Zhu, Renhai Zhong, Zhixian Lin, Jialu Xu, Hao Jiang, Jingfeng Huang, Haifeng Li, and Tao Lin. 2020. "DeepCropMapping: A Multi-Temporal Deep Learning Approach with Improved Spatial Generalizability for Dynamic Corn and Soybean Mapping." *Remote Sensing of Environment* 247:111946. https://doi.org/10.1016/j.rse.2020.111946.

Yan, L., and D. P. Roy. 2015. "Improved Time Series Land Cover Classification by Missing-Observation-Adaptive Nonlinear Dimensionality Reduction." *Remote Sensing of Environment* 158:478–491. https://doi.org/10.1016/j.rse.2014.11.024.

You, Liangzhi, Stanley Wood, Ulrike Wood-Sichra, and Wenbin Wu. 2014. "Generating Global Crop Distribution Maps: From Census to Grid." *Agricultural Systems* 127:53–60. https://doi.org/10.1016/j.agsy.2014.01.002.

Zhai, Yongguang, Zhongyi Qu, and Lei Hao. 2018. "Land Cover Classification Using Integrated Spectral, Temporal, and Spatial Features Derived from Remotely Sensed Images." *Remote Sensing* 10 (3): 383. https://doi.org/10.3390/rs10030383.

Zhang, Chen, Liping Di, Li Lin, Hui Li, Liying Guo, Zhengwei Yang, G. Yu Eugene, Yahui Di, and Anna Yang. 2022. "Towards Automation of in-season Crop Type Mapping Using Spatiotemporal Crop Information and Remote Sensing Data." *Agricultural Systems* 201:103462. https://doi.org/10.1016/j.agsy.2022.103462.

Zhao, Yifan, Weiwei Zhu, Panpan Wei, Peng Fang, Xiwang Zhang, Nana Yan, Wenjun Liu, Hao Zhao, and Qirui Wu. 2022. "Classification of Zambian Grasslands Using Random Forest Feature Importance Selection during the Optimal Phenological Period." *Ecological Indicators* 135:108529. https://doi.org/10.1016/j.ecolind.2021.108529.

Zheng, Baojuan, James B. Campbell, and Kirsten M. de Beurs. 2012. "Remote Sensing of Crop Residue Cover Using Multi-temporal Landsat Imagery." *Remote Sensing of Environment* 117:177–183. https://doi.org/10.1016/j.rse.2011.09.016.

Zhong, Liheng, Peng Gong, and Gregory S. Biging. 2014. "Efficient Corn and Soybean Mapping with Temporal Extendability: A Multi-Year Experiment Using Landsat Imagery." *Remote Sensing of Environment* 140:1–13. https://doi.org/10.1016/j.rse.2013.08.023.